

COMS 4444 Sock Project – Group 4

Rohun Agrawal

October 2026

1 Introduction

This project involves a scenario where roommates share socks from a single drawer. The drawer is initialized with a capacity of C socks and starts full with an equal number of black and white socks. Each day, the n roommates access the drawer in a random order and selects $u \in \{4, 5\}$ socks at random. From these socks, each roommate chooses two to wear and decides whether to return or discard the remaining socks.

Sock shades change gradually with use. White socks begin at shade 255 and decrease by 2 after each wear until reaching 127, while black socks begin at shade 0 and increase by 1 after each wear until reaching 64. Wearing two socks i, j with shades s_i, s_j , costs embarrassment $|s_i - s_j|$ when the difference exceeds 6, and nothing otherwise. When socks are at their terminal shade, they also have a 25% chance of developing a hole after being worn and must then be discarded.

Discarded socks are replaced in packs of six. Whenever six discarded socks of a given color have accumulated, a new pack is purchased for \$10, provided the total spending does not exceed the simulation budget B . If the drawer contains fewer than two socks when a roommate tries to select stocks, that roommate wears no socks and incurs an embarrassment score of 256^2 .

The objective is to design a sock selection and discard strategy that minimizes total embarrassment over T days while utilizing the available budget B . Our group can define the strategy for only the group 4 roommate. Thus, it is possible that we do not have access to strategies from other roommates and our own strategy must take that into consideration. Useful variables to keep track of include S_t , the amount of the budget that has been spent by day t , and D_t , the number of socks our group 4 roommate has chosen to discard up until day t .

2 Initial Strategy

Our first two deliverables combined greedy pair selection with simple rules for deciding which unworn socks to discard. The main intuition was that minimizing embarrassment requires both a good pair today and a drawer that will produce good pairs in future turns. Pair selection addresses the first goal directly, while discarding lets us influence the second by replacing some socks with fresh ones.

2.1 First Deliverable: Closest Pair and Fixed Discard Thresholds

Each turn, we wore the pair with the smallest shade difference:

$$(i, j) = \arg \min_{i < j} |s_i - s_j|.$$

This minimizes the current turn’s embarrassment. With only $u \in \{4, 5\}$ socks drawn, there are just six or ten pairs to compare. Although every pair within six shades incurs zero embarrassment, our initial rule still preferred the smallest gap among those pairs.

For the unworn socks, we used a minimum white shade and a maximum black shade. A white sock below its threshold or a black sock above its threshold was discarded. White socks darken with use and black socks lighten, so these rules remove socks that have moved sufficiently far from their original shade. In terms of the wear age $a(s)$ used in our final strategy, where $a(s) = (255 - s)/2$ for white socks and $a(s) = s$ for black socks, both rules impose an upper age limit on retained socks.

The motivation was to keep the socks of each color concentrated near their fresh shade. A drawer can contain many socks of the same color yet still produce poor matches if their shades are spread across different stages of wear. Replacements arrive as identical fresh socks, so removing older socks could reinforce a population of similar shades and improve future matching opportunities. This benefit extends to every roommate using the shared drawer, even though each player observes only their own handful.

Age was nevertheless only a proxy for usefulness. Two older socks can form an excellent pair, and an isolated fresh sock can be difficult to match. The fixed thresholds offered a simple way to manage future draws without estimating the drawer’s full shade distribution. For a leftover exactly at a threshold, we retained the greedy baseline’s fallback: among such leftovers, consider the one furthest from the worn pair’s mean shade and discard it if the difference exceeds six and the remaining budget can buy a pack.

2.2 Second Deliverable: Wear-Age Tie Breaking and Budget Protection

Our second deliverable retained the fixed discard thresholds and added two refinements. First, when pairs tied for the smallest shade difference, we preferred the pair with the smallest combined wear age $a(s_i) + a(s_j)$. This used equally good immediate matches to influence future sock deterioration. Wearing younger socks could delay losses from holes, since socks at their terminal shade have a 25% chance of developing a hole when worn. The age estimate saturates at 64, so it cannot distinguish how many additional wears terminal-shade socks have survived.

Second, we added a spending check before allowing voluntary discards. Aggressive replacement can maintain closely matched socks while money is available, but each discard contributes to a future pack purchase. Once the budget is exhausted, losses shrink the drawer and may eventually produce sockless turns, each costing 256^2 embarrassment. This makes retaining a usable sock valuable even when its shade is undesirable.

At the start of day t , spending reflects purchases through the preceding day. Using S_{t-1} , we estimated the daily spending rate as

$$\bar{c}_t = \begin{cases} 0, & t = 1, \\ \frac{S_{t-1}}{t-1}, & t > 1. \end{cases}$$

We allowed discards only when extending this rate over the remaining days still left a reserve:

$$B - S_{t-1} - \bar{c}_t(T - t + 1) \geq R_{\text{initial}}, \quad R_{\text{initial}} = \min(C, 30).$$

If this condition failed, we wore the selected pair and returned every unworn sock. With no budget limit, the spending check imposed no restriction.

This check made the discard policy respond to household spending and the remaining horizon. It was a coarse safeguard: purchases happen in batches, and the other roommates may change

their spending behavior. It also allowed all threshold-based discards when the check passed, rather than assigning a specific daily allowance. These limitations motivated the more explicit allocation of discard opportunities in our final strategy.

We explored the fixed shade thresholds through simulation sweeps, comparing embarrassment and spending across candidate settings. The selected thresholds were 240 for white socks and 7 for black socks, corresponding to wear ages of 7.5 and 7. The main development across these deliverables was the progression from controlling shade variation to also considering whether the household could sustain the resulting replacement costs.

3 Final Strategy

Each day, our player wears the drawn pair with the lowest embarrassment and may discard unworn socks so that fresh ones replace them, lowering future embarrassment. Two questions decide the discards.

- (1) **How many:** We estimate how much of the budget is left for us after the other roommates' projected spending, and spread it evenly over the T days (Section 3.2).
- (2) **Which ones:** When the budget is large, discards are plentiful, so a simple rule suffices: discard socks much older than the other socks of their color (Section 3.3.1). When the budget is small, each discard must count, so we discard the socks whose replacement by a fresh sock most lowers expected future embarrassment (Section 3.3.2).

3.1 Notation

Wear age. White socks change by 2 shades per wear and black socks by 1, so to compare socks across colors we measure each sock by how many times it has been worn:

$$a(s) = \begin{cases} \frac{255 - s}{2}, & \text{for white socks,} \\ s, & \text{for black socks.} \end{cases}$$

A fresh sock has age 0, and a sock at its final shade (127 for white, 64 for black) has age 64.

Budget size. Several of our choices depend on how often the budget can replace socks. We measure this with ρ , the average number of times each purchased sock must be worn for the budget to cover all T days. The roommates wear $2nT$ socks in total, and the budget buys $6B/10$ socks, so

$$\rho = \frac{2nT}{6B/10} = \frac{10nT}{3B}.$$

We call the budget *large* when $\rho \leq 3$, that is, when it can replace each sock after at most 3 wears, and *small* otherwise. With no budget limit, $\rho = 0$.

3.2 Set Number of Socks to Discard from Projected Household Spending

We want to discard as many socks as the budget allows without overspending. Since we cannot see the other roommates' spending directly, we

1. keep a reserve R unspent as a safety margin for avoiding overspending,

2. subtract the other roommates' projected spending to obtain our own budget b , and
3. spread b evenly over the T days.

This gives the *allowance*, the most socks we may discard today.

Reserve. The reserve protects against overspending, for example when the other roommates spend more than projected. We set

$$R = \max(30, 0.1B),$$

that is, \$30 or 10% of the budget, whichever is larger (chosen by hyperparameter sweep). When the budget is small, this would leave money unused at the end, so we cap R at the cost of replacing the socks expected to wear out in the remaining days:

$$R \leq 10 \cdot \max\left(1, \left\lceil \frac{2n(T-t+1)}{6L} \right\rceil\right).$$

The remaining $T - t + 1$ days bring $2n(T - t + 1)$ wears, and a sock lasts $L = 68$ wears on average: 64 to reach its final shade, then an expected 4 more before it gets a hole (25% chance per wear). Dividing gives the number of socks that wear out; these are bought in packs of 6 for \$10, and we keep at least one pack. As fewer days remain, the cap falls and releases money for discards.

Our projected budget. Our own budget b is what is left after the reserve and the other roommates' projected spending. During the first W days, there is too little spending history to extrapolate, so we assume every roommate spends equally and set

$$b = \frac{B - R}{n}.$$

Afterward, we estimate the other roommates' spending so far as the household total minus the cost of our own discards (\$10/6 each), and assume they keep spending at the same daily rate until day T :

$$b = \max\left(0, B - R - \max\left(0, S_t - \frac{10D_t}{6}\right) \frac{T}{t}\right).$$

We count our own discards as soon as we choose them, because S_t includes them only once six socks of that color have been discarded.

Pace out spending. Spreading b evenly keeps discards available every day rather than using them up early. Our budget buys

$$q = \frac{6b}{10}$$

socks, so by day t we may have discarded at most qt/T of them. Having already discarded D_t , today we may discard

$$\text{allowance} = \max\left(0, \min\left(A, \left\lfloor \frac{qt}{T} - D_t \right\rfloor\right)\right),$$

where $A = 2$ caps discards per turn. With $u = 4$, this allows discarding both unworn socks. With $u = 5$, it keeps the least worthwhile of the three unworn socks, saving allowance for better discards in later turns.

Other cases where we do not discard. The allowance is also zero in three cases:

- (1) the household is overspending, that is, S_t exceeds $(B - R)t/T$, the amount spent by today under an even schedule, by more than one pack (\$10);
- (2) less than \$10 of the budget remains, so no replacement pack can be bought;
- (3) it is one of the last $\lceil C/(un) \rceil$ days.

The household draws un socks per day, so it takes about $C/(un)$ days to draw through the drawer, and a sock bought later would likely never be worn. With no budget limit, the allowance is A except in case (3).

3.3 Choose Socks to Wear and Discard

We wear a pair with the smallest embarrassment score. When several pairs tie, we wear the one that leaves the most valuable discards unworn (defined below for each budget size), and then the one with the youngest socks, saving older socks for discarding.

3.3.1 Large Budget: Discard Socks Unusually Old for Their Color

A sock much older than the other socks of its color is hard to match, because few of them are within 6 shades of it. We therefore discard such socks, but never nearly new ones. The more discards the budget can afford, the looser both conditions become.

What the budget affords. We measure this with α , the discards per day our budget supports (q/T , Section 3.2) as a fraction of the per-turn maximum A :

$$\alpha = \min\left(1, \frac{q}{AT}\right),$$

with $\alpha = 1$ under no budget limit. Whereas the allowance limits how many socks we discard today, α sets how old a sock must be to be discarded at all.

Threshold 1: Discard unusually old socks. Let μ and σ be the mean and standard deviation of the wear ages of the last $\max(24, 2C)$ socks of one color that our player has drawn. A sock of that color is unusually old when its age exceeds

$$\mu + z\sigma.$$

The multiplier z decreases as α grows: when few discards are affordable, we spend them only on the most extreme socks.

Threshold 2: Safeguard young socks. We do not discard socks at or below a minimum wear age f , to avoid spending on socks that are unusually old for their color but still nearly new. Like z , this minimum is lower when the budget affords more discards. It falls linearly from a_0 when the budget affords almost no discards ($\alpha = 0$) to 1 when it affords the maximum ($\alpha = 1$):

$$f = 1 + (a_0 - 1)(1 - \alpha).$$

We choose $a_0 = 7.5$ for white socks (shade 240) and $a_0 = 7$ for black socks (shade 7) based on hyperparameter sweeping.

Combining the two thresholds. The age above which a sock is unusually old, kept between the minimum f and the largest age 64, is

$$e = \min(64, \max(f, \mu + z\sigma)).$$

An unworn sock is a discard candidate when its wear age $a(s)$ is greater than

$$\tau = \alpha f + (1 - \alpha)e,$$

which lies between f and e and moves from e toward f as the budget affords more discards.

Final discard choice. The threshold τ decides which unworn socks may be discarded, and the allowance decides how many. If there are more candidates than the allowance, we discard those that are the most standard deviations older than the average sock of their color:

$$\text{score}(a) = \frac{a - \mu}{\max(1, \sigma)}.$$

3.3.2 Small Budget: Discard Socks Whose Replacement Lowers Future Expected Embarrassment

When the budget is small, we discard a sock only when replacing it with a fresh sock would lower expected future embarrassment. We

- (1) predict the drawer's contents,
- (2) compute each sock's expected future embarrassment under this prediction, and
- (3) discard the socks whose replacement lowers it most.

Predict drawer. How much embarrassment a sock causes depends on the socks it is drawn with, so we predict what the drawer will contain. We predict the wear ages of the socks of each color based on two sources. The first is what we have seen: $P_{\text{seen}}(a)$, the fraction of the recently drawn socks of that color that have wear age a . The second is what the remaining budget will add: fresh socks, which then age as they are worn. If socks are replaced at a steady rate, so that a sock survives each wear with probability p , their wear ages follow

$$P_{\text{new}}(a) = \begin{cases} p^a(1 - p), & a < 64, \\ p^{64}, & a = 64. \end{cases}$$

The predicted probability that a sock of that color has wear age a is

$$P(a) = (1 - m)P_{\text{seen}}(a) + mP_{\text{new}}(a),$$

where the weight m , between 0 and 1, is the fraction of the drawer that we expect to be replaced soon. A color and a wear age determine a shade (Section 3.1), so this also gives the probability of each shade.

Both m and p depend on r , the number of replacement socks per day that the household can still afford: the remaining budget $B - S_t$ minus the reserve R , converted at 6 socks per \$10 and spread over the $T - t$ remaining days:

$$r = \frac{6}{10} \cdot \frac{\max(0, B - S_t - R)}{T - t}.$$

Unlike q/T in Section 3.2, which counts only what our own budget affords, r counts every replacement, whoever discards. The larger r is, the more the drawer will be refreshed, so the larger m is and the lower p is.

Expected future embarrassment of a sock. Let $V(s)$ be the expected embarrassment of a future turn in which we draw the sock of shade s together with $u - 1$ other socks, each drawn independently from the predicted drawer and equally likely to be white or black. Let G be the smallest shade gap between any two socks in the draw, with gaps above 64 counted as 64. Since we wear the closest pair, the embarrassment is G when $G \geq 7$ and zero otherwise, so

$$V(s) = \mathbb{E}[G \mathbf{1}\{G \geq 7\}] = 7 P(G \geq 7) + \sum_{g=8}^{64} P(G \geq g).$$

We compute each $P(G \geq g)$ exactly, rather than by sampling, with a dynamic program over the sorted shades. It sums the probability of every placement of the other socks below and above s such that all neighboring shades are at least g apart.

Final discard choice. We rank all socks by the reduction in expected future embarrassment

$$V(s) - V(\text{fresh})$$

and discard the ones with the largest reductions.

4 Results

4.1 Experimental Setup and Scoring

The tournament contained 81,216 simulations: 27,072 configurations, each repeated with three seeds. It included nine groups, households of 1, 2, 5, 9, 18, and 36 roommates, both $u = 4$ and $u = 5$, four drawer multipliers, and twelve budget and duration settings. The settings ranged from no replacement money to an unlimited budget, with durations from one to ten years. The raw file contains 578,016 roommate records and no recorded player faults. ~~These results evaluate the tournament submission; they do not separately measure the contributions of our earlier strategies or individual components of the final strategy.~~ Section 4.7 adds one controlled experiment: we replay the tournament households in which Group 4 held most of the seats with our voluntary discards switched off.

For each group present in a simulation, we first average its roommates' outcomes. Unless stated otherwise, we then give each simulation equal weight. Thus, eight copies of a group in one household do not count as eight independent experiments. Each group appears in 29,376 simulations. The supplied standings instead average individual roommate rows, giving more weight to runs with more copies of a group. Appendix A reports both conventions. All plots are descriptive averages over the tested settings. With only three repetitions per configuration, we do not interpret small differences as precise rankings of strategy quality.

To separate matching quality from shortages, let E be a roommate's total embarrassment and N_0 its number of sockless days. We calculate

$$E_{\text{match}} = E - 256^2 N_0.$$

The daily score is E/T , and the mismatch component is E_{match}/T . We also report mismatch embarrassment per dressed day by dividing the sum of mismatch embarrassment by the sum of $T - N_0$ across observations. This avoids treating a day without socks as successful matching. Household spending and the first day on which a replacement purchase failed are counted once per simulation. A failed purchase does not necessarily imply a subsequent sockless turn: enough socks may remain to finish the run.

4.2 What We Tested Before the Tournament

Our development tests covered a narrower range of households than the tournament. The future-hand discard rule (Section 3.3.2) was accepted on four roommates sharing $C = 40$ socks for 1,080 days with $B = \$800$ ($\rho = 18$); over 20 seeds it lowered mean embarrassment from 6.96 to 0.18 per roommate-day. The final changes to the budget controller were tested in every two- and four-player lineup of the nine groups with $C = 40$, $u = 4$, 1,080 days and $B = \$7,490$ ($\rho = 1.9$, a large budget). Over 8,568 simulations they lowered Group 4’s mean embarrassment by 20.3% (95% interval 6.4–32.9%), leaving Group 4 with the lowest field mean, statistically tied with Group 8. No test had more than nine roommates in a household.

Every finite budget in the tournament had $\rho \geq 3.6$, so the large-budget rule (Section 3.3.1) ran only in the unlimited-budget games, and the small-budget rule decided every other game, including households of 18 and 36 roommates.

4.3 Overall Performance: Shortages Dominate the Score

Figure 1 and Table 1 show that Group 4 was near the bottom under equal simulation weighting, with mean embarrassment of 923.63 per roommate-day. Its median was only 0.035, illustrating that most runs had very small scores while a minority incurred severe penalties. Group 4 experienced at least one sockless turn in 5.33% of its simulations, and sockless days accounted for 99.95% of its mean score. Preventing relatively rare availability failures would therefore have a much larger effect on this aggregate score than a comparable improvement in ordinary matching. **Ranked within each of the 1,152 configurations, Group 4 placed first in 325 and last in 249.**

This does not mean our matching policy was best. Its mismatch embarrassment per dressed day was 0.689, the highest among the nine groups, compared with 0.471 for Group 8. However, the scale of these differences is small compared with the shortage penalty. ~~Both components matter: a policy can match well while socks are available and still perform poorly over the full horizon.~~

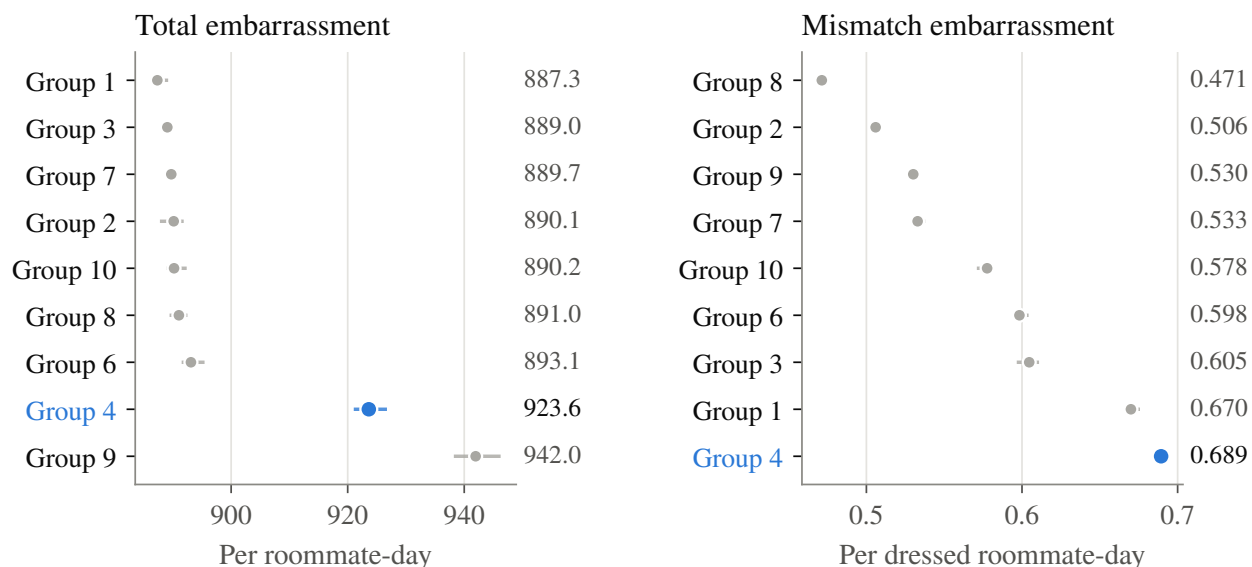


Figure 1: Total embarrassment with equal simulation weighting (left), and mismatch embarrassment per dressed roommate-day (right), with groups sorted from best to worst in each panel. The right panel pools embarrassment and dressed days before division, so longer runs contribute more dressed days. ~~Error bars span~~ Horizontal lines span the three seed-specific aggregate results, not confidence intervals. Group 4 is our submission.

Group	Mean daily score	Median daily score	Mismatch/dressed day	Sockless runs (%)
1	887.33	0.072	0.670	5.05
2	890.14	0.019	0.506	5.05
3	889.05	0.026	0.605	5.00
4	923.63	0.035	0.689	5.33
6	893.10	0.025	0.598	5.14
7	889.73	0.011	0.533	5.33
8	891.03	0.021	0.471	5.22
9	941.96	0.028	0.530	5.57
10	890.20	0.026	0.578	5.07

Table 1: Each mean and median uses one average outcome per group per simulation. A sockless run has at least one sockless day for a member of that group.

4.4 Losses Come from Households Dominated by Group 4

Figure 2 breaks the result down by household. Group 4 had the best mean rank of the nine groups in two-person mixed households (2.62) and the second best alone (1.84), where it placed first in 73 of 96 configurations. In mixed households of 5 to 36 roommates its mean rank stayed between 4.2 and 5.0. It fell to between 7.0 and 7.8, last among the nine groups, in the households where it held most of the seats: nine, 18 or 36 copies of our player, or eight Group 4 players with one other group. Those four household types account for 33.9 of the 36.3-point gap between Group 4 and Group 1, the leader under equal simulation weighting; the 36-roommate all-Group-4 households alone account for 17.0. The one other weak case is a single Group 4 player among eight copies of another group (mean rank 6.08).

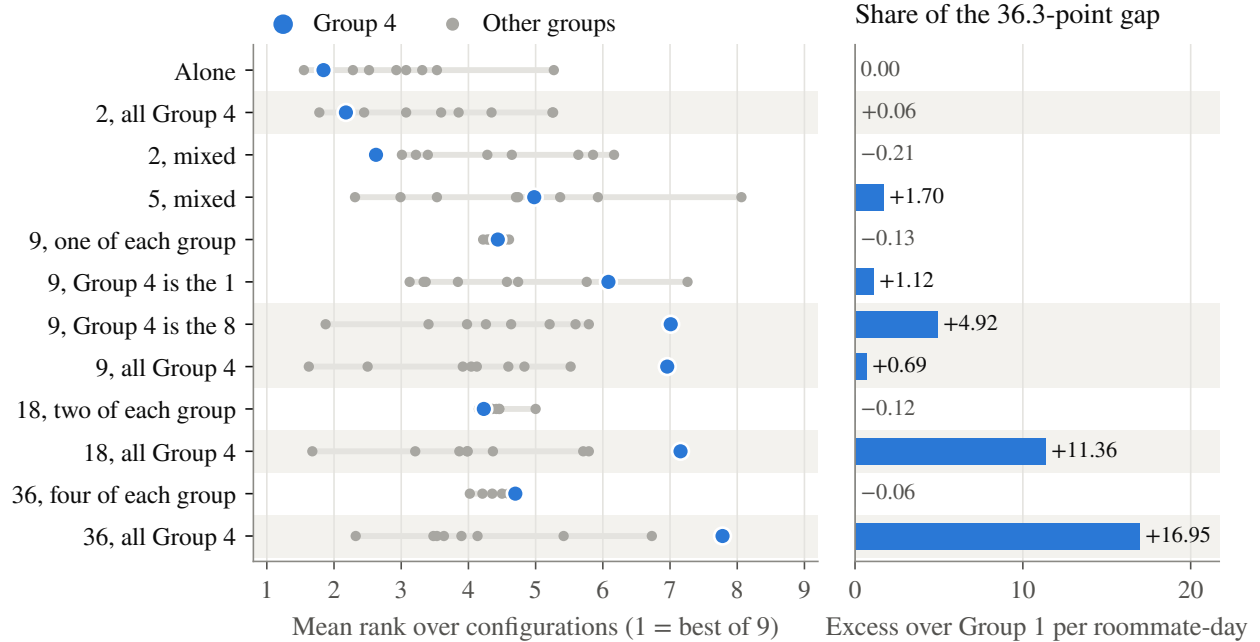


Figure 2: Left: Group 4’s mean rank in each household type, against the other eight groups, over the 96 configurations of that household (budgets, durations, draw sizes and drawer multipliers). Right: each household type’s contribution to the difference between Group 4’s and Group 1’s equal-simulation mean scores. Shaded rows are households in which Group 4 holds most of the seats.

4.5 Budget, Duration, and Household Size

Figure 3 restricts attention to households containing only Group 4 players. Each cell averages both draw sizes, all four drawer multipliers, and the three repetitions. Group 4 had no sockless days in any single-roommate run and averaged only 0.010 embarrassment per day in that setting. Larger households were much more vulnerable, but the household budget B was not increased with n . Increasing n consequently reduced both budget per roommate-year and, at a fixed drawer multiplier and draw size, initial socks per roommate. The comparison does not isolate a cost of coordination.

Duration also matters. With nine Group 4 roommates and $B = 150$, one-year simulations averaged 0.496 embarrassment per day and had no sockless runs. Extending the same budget to two years increased the mean to 4,806.05, with sockless days in 41.7% of runs. Even the largest finite budget, $B = 1500$, did not guarantee survival: over ten years, all 36-roommate homogeneous Group 4 runs had sockless days. Conversely, all homogeneous Group 4 runs with unlimited replacement money had zero embarrassment. ~~The main failures therefore occurred under resource limits, although low mismatch scores under unlimited replacement do not establish that the same replacement behavior is economical.~~ Across all households with unlimited money, where the large-budget rule ran, Group 4 placed first in 82 of 96 configurations.

Mismatch embarrassment per roommate-day												
Roommates	1	0.01	0.01	0.00	0.00	0.07	0.00	0.01	0.00	0.00	0.00	0.00
	2	0.38	0.09	0.02	0.02	0.18	0.03	0.04	0.01	0.00	0.02	0.01
	9	0.10	1.26	0.50	1.43	5.75	2.15	2.50	0.68	0.20	1.05	0.60
	18	0.08	0.75	1.01	1.66	3.10	2.18	3.41	2.41	1.28	4.63	3.46
	36	0.05	0.62	0.68	0.97	1.36	1.15	1.84	1.56	1.22	2.78	2.48
		\$0	\$150	\$150	\$300	\$1,500	\$500	\$1,000	\$500	\$300	\$1,500	\$1,000
		1 yr	2 yr	1 yr	2 yr	10 yr	3 yr	5 yr	2 yr	1 yr	5 yr	3 yr
												Unlimited
												1 yr
Simulations with any sockless day (%)												
Roommates	1	0	0	0	0	0	0	0	0	0	0	0
	2	12	0	0	0	0	0	0	0	0	0	0
	9	38	42	0	0	38	0	0	0	0	0	0
	18	38	62	25	50	100	75	75	38	25	25	0
	36	50	75	50	75	100	75	75	50	25	75	75
		\$0	\$150	\$150	\$300	\$1,500	\$500	\$1,000	\$500	\$300	\$1,500	\$1,000
		1 yr	2 yr	1 yr	2 yr	10 yr	3 yr	5 yr	2 yr	1 yr	5 yr	3 yr
												Unlimited
												1 yr

Figure 3: Homogeneous Group 4 households. Top: mismatch embarrassment per roommate-day. Bottom: percentage of simulations with any sockless days. Columns are ordered by annual household budget, retaining the actual budget and duration. Each cell contains 24 simulations; household size changes the resources available per roommate.

4.6 Replacement Pressure and Budget Exhaustion

Figure 4 compares homogeneous households using $\rho = 10nT/(3B)$, the replacement pressure defined in Section 3.1. Higher ρ means each purchased sock must support more wears. The figure includes only positive finite budgets. Outcomes at each pressure value are averaged over the compatible tested settings, so duration and initial stock still matter within this comparison.

Across all homogeneous Group 4 runs, a replacement purchase failed in 30.9%, whereas sockless days occurred in 22.8%. In 8.1% of runs, purchases failed but the remaining drawer lasted through the horizon. ~~Exhaustion is therefore a useful warning rather than an interchangeable measure of failure.~~ Its timing matters as well: an early failed purchase leaves more time for holes and discards to deplete the remaining stock. ~~The high-pressure portion of Figure 4 shows earlier purchase failures and a larger fraction of sockless days. Other groups also struggled in this region, indicating that some tested budgets were broadly difficult, while Group 4's additional losses show that its safeguards did not fully control this risk.~~ Group 4 households were the first to run out of money as pressure rose. At $\rho = 72$ and $\rho = 86$, their budgets ran out in 53% and 75% of runs, against at most 28% and 38% for any other group. Above $\rho \approx 100$, every group ran out of money in most runs, and Group 4 spent the most roommate-days without socks at every pressure level but one.

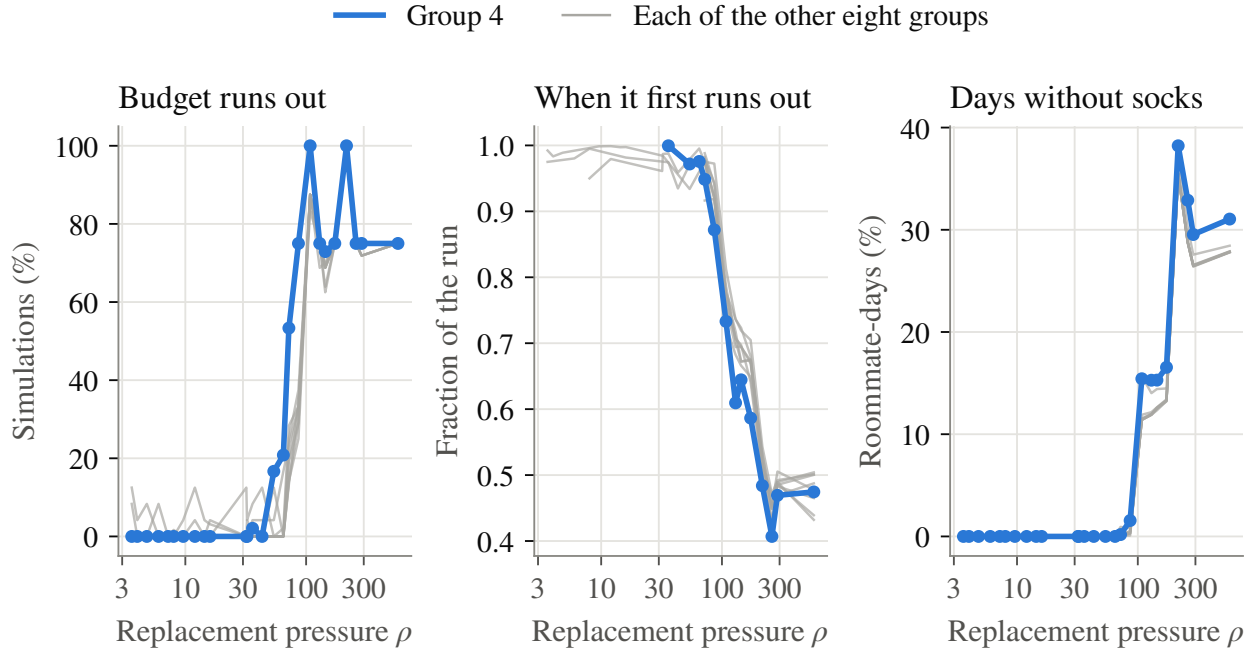


Figure 4: Positive finite budgets in homogeneous households. The panels show the frequency of failed replacement purchases, their average timing conditional on failure, and the percentage of roommate-days without socks. ~~The comparison line is the median of the eight other groups' averages at each ρ .~~ Gray lines show each of the other eight groups. The horizontal axis is logarithmic; connected points summarize tested settings rather than a fitted response curve.

4.7 Voluntary Discards Caused the Excess Shortages

To test whether our discards caused these failures, we replayed every finite-budget tournament simulation in which Group 4 held most of the seats, and the nine-person households with a single

Group 4 player: 5,280 simulations in total. Each replay kept the roster, seat order, drawer, budget, duration and seed. The only change was that Group 4’s discard allowance (Section 3.2) was always zero; worn-out socks with holes were still replaced. Forty randomly chosen simulations replayed with the unmodified player reproduced every roommate’s tournament score exactly.

Table 2 and Figure 5 show the effect. In the 18-roommate all-Group-4 households, mean embarrassment fell from 7,221 to 5,923 per roommate-day and the budget ran out in 45% of runs instead of 69%. Among the nine groups’ means in those households, Group 4 moved from last to third. With nine copies, or eight Group 4 players and one other group, Group 4 moved from eighth to sixth and from eighth to second. In the two-person households the budget covered both holes and discards, and there the discards helped: switching them off raised mean embarrassment from 49.7 to 50.6.

Replacing the replayed simulations in the standings, with every roommate’s score taken from the replay, lowers Group 4’s mean from 923.63 to 887.92. That would rank second, behind Group 1 at 886.58; the other groups’ means also fall slightly because they share some of the replayed households. Almost all of the gap in Figure 2 therefore comes from voluntary discards in households whose budget could not pay for them.

Household	Simulations	Embarrassment / day		Budget ran out (%)		Field median
		Submitted	No discards	Submitted	No discards	
2, all Group 4	264	49.7	50.6	1.1	1.1	47.7
9, all Group 4	264	1,129.9	1,071.3	23.1	12.1	1,067.3
9, Group 4 is the 8	2,112	1,171.6	1,103.9	24.3	14.1	1,118.5
18, all Group 4	264	7,221.3	5,922.9	68.6	45.5	5,945.9
36, all Group 4	264	16,591.1	14,660.3	75.8	75.0	14,672.6
9, Group 4 is the 1	2,112	1,123.3	1,105.2	17.7	15.7	1,119.5

Table 2: Group 4’s mean embarrassment per roommate-day and the share of simulations in which the household budget ran out, in the tournament and in the replay without voluntary discards. All finite budgets. The field median is the median of the other eight groups’ means in the same household type and role.

The budgets in these households could not cover holes, let alone voluntary discards. A sock lasts about 68 wears (Section 3.2), so n roommates wear out about $2nT/68$ socks over T days. For 18 roommates over five years that is about 950 socks, or \$1,590 of six-packs, against a \$1,000 budget. Our reserve held back at most $\max(30, 0.1B)$, and each Group 4 player treated the rest of its share as available for voluntary discards. Every such discard spent money that hole replacements later needed, so the household ran out of money sooner, and after that each hole removed a sock from the drawer for good.

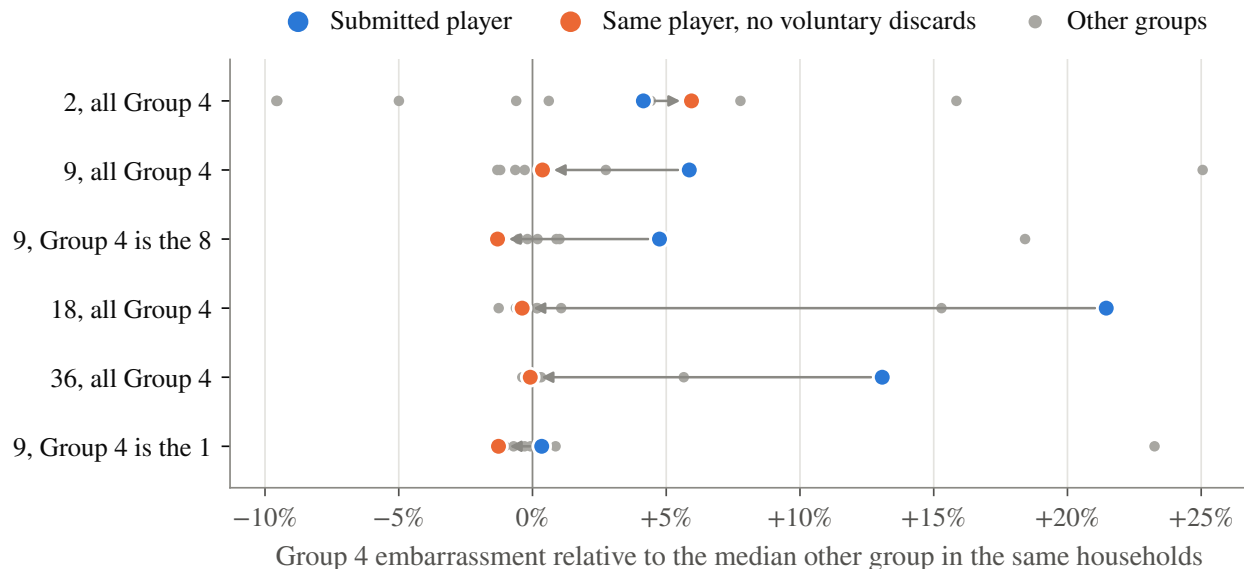


Figure 5: Group 4's mean embarrassment in the tournament (blue) and in the replay without voluntary discards (orange), relative to the median of the other eight groups in the same household type and role, over all finite budgets. Gray points are the other eight groups.

4.8 The Influence of Other Roommates

Figure 6 compares Group 4 with each other group under the same budget, duration, draw-size, and drawer-multiplier grid. In two-person households, its mean score ranged from 30.40 with Group 2 to 63.91 with Group 10. Yet every partner produced the same 1.04% frequency of runs with Group 4 sockless days. Differences in total score can therefore reflect the severity and duration of shortages, even when their frequency is unchanged.

In nine-person households, a single Group 4 player among eight Group 9 players averaged 1,223.54 embarrassment per day, compared with 991.84 among eight Group 3 players. When eight players used Group 4 and only one used another policy, Group 4 averaged 1,353.56 with a Group 9 neighbor. These results show sensitivity to household composition, but the best neighbor depends on the metric: among eight Group 7 players, Group 4's mismatch component was only 0.418 per day, compared with 1.215 among eight Group 1 players. ~~A neighbor that supports close matching need not minimize shortage risk.~~

The raw file records household spending, not individual discard decisions or individual expenditure. We therefore cannot identify which roommate caused a particular purchase failure from the tournament records alone; the replay in Section 4.7 addresses this for households dominated by Group 4. Appendix B extends this comparison to changes in the outcomes of other groups when Group 4 enters a five-person household.

Group 4 total embarrassment per roommate-day								
1 Group 4 + 1 other	46	30	39	48	39	39	38	64
1 Group 4 + 8 other	995	1,007	992	1,003	995	1,022	1,224	1,001
8 Group 4 + 1 other	1,032	1,037	1,034	1,015	1,032	1,054	1,354	1,035
	Group 1	Group 2	Group 3	Group 6	Group 7	Group 8	Group 9	Group 10
Group 4 mismatch embarrassment per roommate-day								
1 Group 4 + 1 other	0.10	0.06	0.04	0.12	0.06	0.07	0.08	0.06
1 Group 4 + 8 other	1.22	0.68	0.77	0.80	0.42	0.47	0.61	0.81
8 Group 4 + 1 other	1.75	1.20	1.27	1.29	1.05	1.15	1.12	1.26
	Group 1	Group 2	Group 3	Group 6	Group 7	Group 8	Group 9	Group 10
Simulations with any Group 4 sockless day (%)								
1 Group 4 + 1 other	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
1 Group 4 + 8 other	8.3	8.3	8.3	8.7	11.5	10.8	11.8	8.3
8 Group 4 + 1 other	9.0	9.0	10.4	10.1	10.1	10.8	12.5	9.4
	Group 1	Group 2	Group 3	Group 6	Group 7	Group 8	Group 9	Group 10
The other group in the household								

Figure 6: Group 4 outcomes by the other group and Group 4's role in the household. Each cell averages 288 simulations covering the same resource grid. In the majority case, scores first average the eight Group 4 players within a simulation; a sockless run requires at least one of those players to go without socks.

4.9 Initial Drawer Capacity Can Delay or Prevent Shortages

Figure 7 varies drawer capacity within fixed household types and budget-duration settings. For 36 Group 4 roommates with $B = 300$ over one year, doubling the base drawer reduced mean embarrassment from 12,898.96 to 1.85 and eliminated sockless days in the tested runs. Initial capacity can consequently determine whether a household survives a short horizon even when replacement money per roommate is small.

The protection was incomplete over longer horizons. With 36 Group 4 roommates and $B = 1500$ over ten years, increasing the drawer multiplier from 1 to 10 reduced sockless roommate-days from 73.0% to 36.5%, but did not remove shortages. A large initial drawer is a finite buffer, so its contribution becomes smaller relative to total wear as T grows. The mixed 36-person households showed a similar pattern.

More capacity also did not improve ordinary matching monotonically. In nine-person homogeneous households with $B = 300$ over one year, there were no sockless days at any drawer multiplier, but daily embarrassment increased from 0.139 at multiplier 1 to 0.373 at multiplier 2 before falling to 0.041 at multiplier 10. Capacity changes the evolving shade distribution as well as the available stock; the aggregate records cannot determine the exact mechanism behind this pattern.

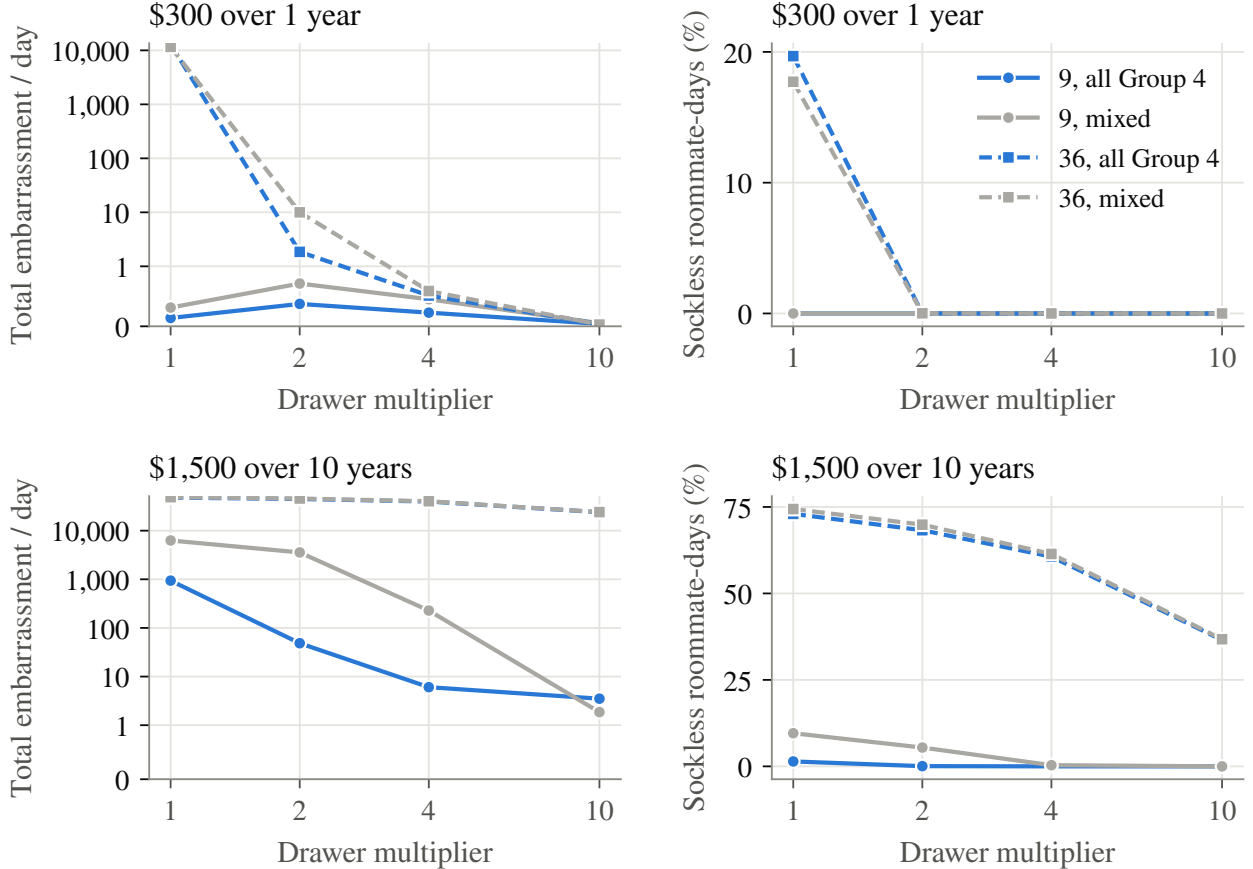


Figure 7: Group 4 in homogeneous and mixed households of 9 and 36 roommates, holding budget and duration fixed within each row. Each point averages both draw sizes and three repetitions. Mixed households contain one or four players from each group, respectively. Total-score axes are logarithmic above 1 and linear near zero; sockless percentages use linear axes.

4.10 Drawing Four Versus Five Socks

We matched the $u = 4$ and $u = 5$ runs by roster, budget, duration, drawer multiplier, and seed, giving 14,688 paired comparisons for Group 4. Figure 8 shows an improvement at every household size. Across these pairs, mean daily embarrassment fell from 1,071.10 to 776.16, a reduction of 27.5%. The mismatch component fell from 0.751 to 0.084 per roommate-day, and sockless roommate-days fell from 1.63% to 1.18%. Thus, the improvement included both better matching and greater availability. **Relative to the field, Group 4 also did better with five socks: its mean rank within configurations was 4.44 with $u = 5$, against 5.56 with $u = 4$.**

These comparisons evaluate the tested five-sock setup as a whole. Its base drawer was also larger: for example, with nine roommates it contained 60 socks instead of 48, and with 36 roommates it contained 192 instead of 156. There were no paired runs holding the actual drawer capacity C constant while changing u . The extra candidate sock increases the number of possible pairs from six to ten, but these data cannot separate that advantage from the benefit of the larger drawer.

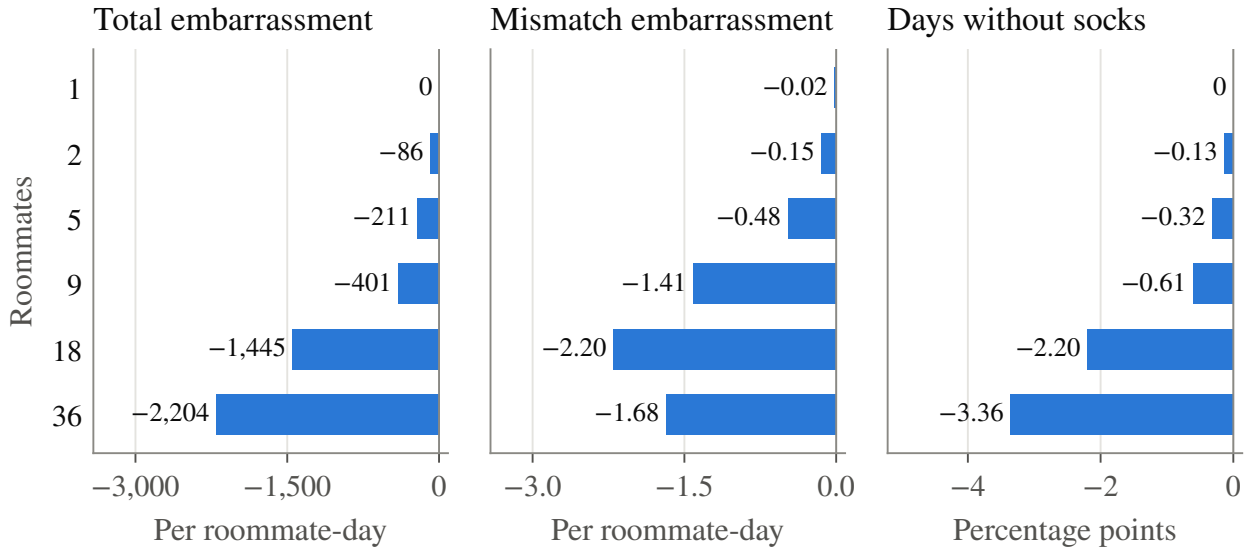


Figure 8: Mean paired changes for Group 4 from the four-sock to the five-sock setup, grouped by household size. Negative values indicate improvements. Each comparison holds roster, budget, duration, drawer multiplier, and seed fixed, but actual drawer capacity increases with draw size.

5 Discussion

Our results demonstrate the central tradeoff in the Socks problem: reducing short-term embarrassment through aggressive sock replacement must be balanced against the long-term cost of maintaining the shared drawer. Discarding an aged or poorly matched sock can improve the quality of future draws, but every six discarded socks of the same colour triggers cost of a replacement pack. If the household exhausts its budget, discarded socks are no longer replenished, which can eventually leave roommates without enough socks to wear and incur a substantially larger embarrassment penalty. Our strategy therefore cannot optimize each day’s hand independently; it must consider how present decisions affect both future pair quality and the remaining household budget.

A significant limitation of our approach is the incomplete information available to each player. A roommate observes only the socks drawn on their own turn and cannot track which socks other

roommates have worn or the current number of discarded socks awaiting replenishment. Although the simulator provides the total amount spent, this reveals a purchase only after a replacement pack has already been triggered. Consequently, our strategy must estimate the state of the shared drawer and the effect of its own discard decisions rather than calculate them exactly. The neighbor comparisons demonstrate that household composition matters, but the raw records do not reveal whether a particular difference arose from discarding, wear choices, or subsequent holes. Daily observations would be needed to attribute those effects to a specific mechanism.

Budget management introduces an additional source of uncertainty. Our player attempts to pace discretionary discards according to the remaining budget and simulation horizon, but it cannot directly observe how close each colour is to triggering its next six-sock replacement pack. Moreover, worn-out socks that develop holes also require replacements, so not all of the budget can safely be allocated to voluntary discarding. The distinction between failed purchases and sockless turns is important: some households finished after replacement became unaffordable, while others lost much of their remaining stock. Protecting enough socks to reach day T , rather than only controlling spending, is the more direct objective. The fact that 99.95% of Group 4’s aggregate score came from sockless penalties makes this the main priority for improving tournament performance.

There are also limitations to our pair-selection and discard heuristics. White and black socks age at different rates; as a result, shade alone does not provide a uniform measure of remaining sock life. Our later strategy accounts for this distinction for estimating wear, but decisions about whether an unused sock should be preserved or discarded still depend on predictions about future random draws. A sock that is a poor match today may form a useful pair later, while discarding it may alter both the drawer distribution and future replacement timing. The optimal decision therefore depends on a longer horizon that cannot be captured perfectly in a local heuristic.

The household-size comparison needs to be interpreted in terms of resources. The tournament held total budget fixed across household sizes, so large households had less replacement money per person. Their initial stocks per person also differed. Strong single-roommate results and weak large-household results therefore do not by themselves establish that pooling or coordination caused the deterioration. The capacity experiments show that increasing initial stock can eliminate shortages in some short runs, while even a tenfold drawer increase can fail over a long horizon. A controlled pooling comparison should hold per-person resources and duration constant.

6 Future Work

A natural extension of our strategy would be to improve its model of the unobserved drawer state. Rather than relying primarily on recent draws, a future implementation could maintain probabilistic estimates of the number and age distribution of white and black socks. Such a model would not eliminate the partial-observability problem, but it could provide a more principled basis for predicting the value of keeping or replacing a sock.

Budget management could similarly be more probabilistic. Because purchases occur in batches of six discarded socks of the same colour, the marginal cost of a discard depends heavily on the unknown number of previously discarded socks. A future strategy could maintain an estimated distribution over this hidden count for each colour and incorporate the expected replacement cost into its discard decision. It could also reserve an estimated proportion of the remaining budget for unavoidable replacements caused by holes, rather than treating the remaining budget as entirely available for strategic discards.

Further experiments should isolate the effect of drawing five socks rather than four while holding the actual drawer capacity C fixed. The tournament’s five-sock setup improved both matching

and availability, but also started with more socks. A fixed-capacity comparison across several budgets and household sizes would establish how much of the observed improvement comes from the additional candidate sock, which increases the number of possible pairs from six to ten, and how much comes from the larger initial stock.

Finally, the relationship between pooled and individual sock management warrants more controlled experimentation. The results indicate our player performs particularly well with one roommate but scales poorly in some larger households. Future experiments could compare an n -roommate shared drawer against n independent one-roommate simulations while keeping the total number of socks, per-person budget, and simulation length comparable. This would separate the benefit of having a larger pool of potential matches from the costs of incomplete information and independent decision making. Additional experiments with mixtures of different player strategies could further determine whether the poor scaling observed results from pooling itself or from identical agents making similar discard decisions.

A Sensitivity to Aggregation

Table 3 contrasts equal simulation weighting with equal roommate-record weighting. The latter reproduces the mean scores in the supplied overall standings. It emphasizes configurations with many copies of a group, including the large homogeneous households that often ran short of socks. Group 4 ranks eighth under equal simulation weighting and ninth under equal roommate weighting. Neither convention changes the conclusion that its shortage control needs improvement. Since every configuration has three repetitions, equal configuration weighting after averaging seeds is identical to equal simulation weighting.

The broad conclusion also holds for each seed separately. Group 4’s equally weighted mean scores were 926.44, 923.09, and 921.34 for seeds 1, 2, and 3, respectively. It ranked eighth in each case. Several other groups had overlapping ranges of seed-specific means, so their smaller overall differences should be interpreted cautiously.

Group	Equal simulation weight		Equal roommate weight	
	Mean daily score	Rank	Mean daily score	Rank
1	887.33	1	3462.11	5
2	890.14	4	3461.12	4
3	889.05	2	3451.25	2
4	923.63	8	3856.68	9
6	893.10	7	3477.23	7
7	889.73	3	3447.41	1
8	891.03	6	3463.61	6
9	941.96	9	3724.65	8
10	890.20	5	3460.55	3

Table 3: Sensitivity of overall scores and ranks to the unit of aggregation.

B Changes in Other Groups' Outcomes

To examine Group 4's association with outcomes for its neighbors, we matched five-person rosters sharing four groups and differing only in the fifth. For each other group, we compared replacing that group with Group 4 while holding the draw size, drawer multiplier, budget, duration, and seed fixed. There were 10,080 comparisons per replaced group. Figure 9 reports the change in the average score of the four groups retained in both rosters, so the comparison does not include the incoming player's own score.

The changes were small relative to the total scores in five-person households. Replacing Group 1 with Group 4 increased the other four groups' total embarrassment by 1.70 per roommate-day but reduced their mismatch component by 0.080. Replacing Group 2 reduced their total score by 0.159 while increasing the mismatch component by 0.067. These examples again show that matching and sock availability can move in opposite directions. Household spending also changed differently by replaced group: replacing Group 1 reduced annual spending by \$12.28 on average, whereas replacing Group 2 increased it by \$47.94.

These are comparisons within the tested roster and resource grid, not evidence that Group 4 universally helps or harms its neighbors. Matched seeds do not preserve identical random trajectories once sock choices and replacements diverge. Individual discard histories and daily drawer contents would be needed to identify the mechanisms behind the changes.

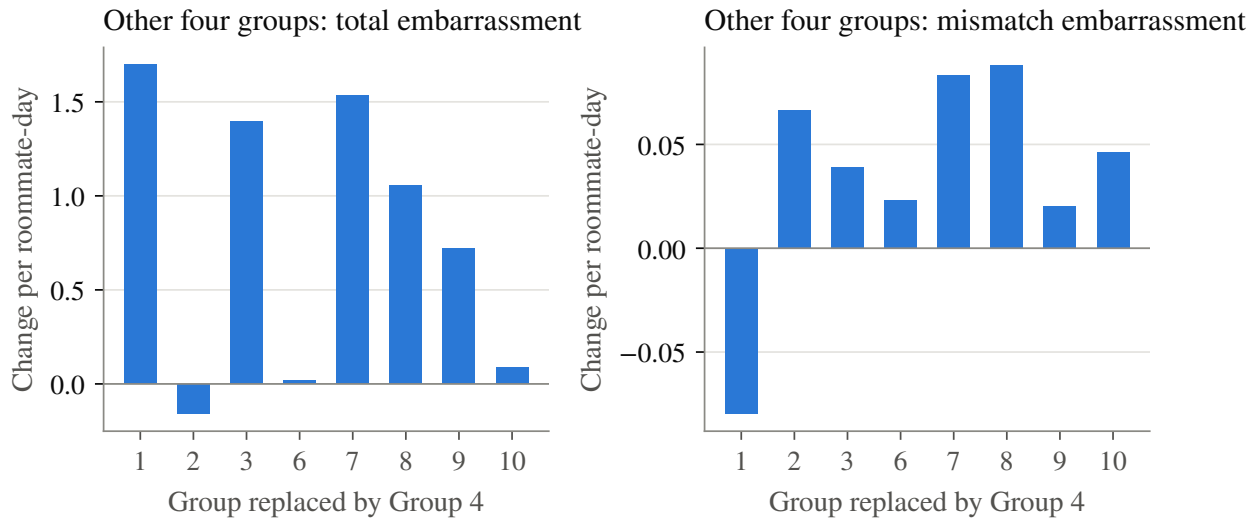


Figure 9: Changes for the four retained groups when the fifth roommate's strategy is replaced by Group 4. Each bar averages 10,080 matched comparisons. Positive values mean greater embarrassment for the retained groups.